live_decisions: Detection Metrics Report

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The detector ranks attacks well but acts on too few of them: ROC-AUC is 0.814 and PR-AUC (average precision) is 0.872 on 140 decisions, yet the block/quarantine actions caught only 47 of 90 attacks (recall 52.2%) with zero false positives.

Confusion matrix and threshold metrics

Treating `block` and `quarantine` as "attack" and `forward` as "benign", the system made no false alarms but missed 43 of 90 attacks.

	Predicted attack	Predicted benign	Total
Actual attack	TP = 47	FN = 43	90
Actual benign	FP = 0	TN = 50	50
Total	47	93	140

Metric	Value	Meaning
Precision	1.000	47 / (47 + 0): every flagged entity was a real attack
Recall (TPR)	0.522	47 / 90: about half of attacks were stopped
Specificity (TNR)	1.000	50 / 50: no benign traffic was blocked
False positive rate	0.000	0 / 50
False negative rate	0.478	43 / 90
F1	0.686	Harmonic mean of precision and recall
Accuracy	0.693	97 / 140 correct; inflated by the 64% attack share
Balanced accuracy	0.761	Mean of recall and specificity
MCC	0.530	Correlation between decision and truth

ROC-AUC and PR-AUC

Both are computed from the continuous `score` column, not from the block/forward action, with attack as the positive class.

Metric	Value	95% bootstrap CI	No-skill baseline
ROC-AUC	0.814	0.760 to 0.866	0.500
PR-AUC (average precision)	0.872	0.819 to 0.914	0.643 (attack prevalence)

ROC-AUC of 0.814 means a random attack outscores a random benign row 81% of the time. PR-AUC of 0.872 is well above the 0.643 baseline, so flagged rows are mostly real attacks even at higher recall. The trapezoid PR-AUC is 0.929, but it interpolates linearly between points and is optimistic, so use average precision.

The curves are step-shaped because `score` has only 15 distinct values and 80 of 140 rows share 0.0883, so the AUCs depend heavily on how those ties are broken. The ROC and PR curve image is attached in chat as `roc_pr_curves.png`.

Where the misses are

Slowloris causes 36 of the 43 misses; floods are mostly caught, and benign traffic is never blocked. Every `mechanism` value is `limit_conn`, so there is nothing to split there.

Truth / profile	Rows	Blocked or quarantined	Forwarded	Recall	Score range
Attack / flood	36	29	7	0.806	0.113 to 0.761
Attack / slowloris	54	18	36	0.333	0.088 to 1.000
Benign / benign	29	0	29	n/a	0.088
Benign / rudy	18	0	18	n/a	0.088 to 0.105
Benign / flood-like	3	0	3	n/a	0.088

Attack type vs benign	ROC-AUC	PR-AUC
Flood vs benign	1.000	1.000
Slowloris vs benign	0.690	0.710

The lowest flood score (0.113) is above the highest benign score (0.105), so a lower threshold would separate floods perfectly. Slowloris is the problem: 36 forwarded slowloris rows score 0.088 to 0.133, the same as benign traffic. The 18 quarantined ones are exactly those with `age` of 30.6 or more, which suggests the stall rule only fires after about 30 units. Benign `rudy` rows with age up to 51.8 stay at a score near 0.088, so age alone does not explain the score.

Sensitivity checks

Lowering the action threshold from 0.761 to 0.113 would catch 8 more attacks (55 TP) with still no false positives.

Score threshold (attack if score >=)	TP	FP	FN	TN	Precision	Recall	FPR
0.0883 (everything)	90	50	0	0	0.643	1.000	1.000
0.0888	58	2	32	48	0.967	0.644	0.040
0.1048	55	1	35	49	0.982	0.611	0.020
0.1129 (best with no FP)	55	0	35	50	1.000	0.611	0.000
0.7612 (current actions)	47	0	43	50	1.000	0.522	0.000
0.8703	18	0	72	50	1.000	0.200	0.000

Subset	n	TP	FP	FN	TN	ROC-AUC	PR-AUC
All decisions	140	47	0	43	50	0.814	0.872
`final = True` rows only	53	28	0	1	24	1.000	1.000
Last decision per entity	74	46	0	14	14	0.902	0.968

The `final = True` result is not a fair estimate: it keeps 28 blocked attacks and 24 benign rows and drops almost every miss, so its perfect AUC reflects how the rows were selected. I assumed `final` marks a settled decision, which the file does not define. The per-entity view is the gentler alternative: repeated rows from the same IP are collapsed, which lifts ROC-AUC to 0.902 but still leaves 14 attack IPs undetected.

Caveats and next steps

The fastest gain is slowloris detection, because its scores are indistinguishable from benign traffic below an `age` of about 30.

- Lower the block threshold to about 0.113: it recovers 8 floods with no false positives on this data, but check it on fresh traffic first since 0.105 and 0.113 sit very close.
- Give the model a slowloris-specific signal (connection hold time or partial-request count); the current score gives it an ROC-AUC of only 0.690 against benign.
- Report metrics per entity as well as per row: the 140 rows come from 74 IPs, and some IPs appear up to 9 times, so rows are not independent.
- Read the confidence intervals, not just the point values: on 140 rows the ROC-AUC could plausibly be anywhere from 0.76 to 0.87.
- Tell me what `final` means if my assumption is wrong, and I will redo the subset analysis.